

**SIMATS ENGINEERING
ASSESSMENT TOOLS**

COURSE NAME: Theory of Computation

COURSE CODE: CSA1304

COURSE OUTCOME COVERED

CO3: Construct regular expressions and use the pumping lemma and closure properties to analyze regular languages. (BL:3)

Assessment Tool Weightage: 40%

QUESTIONS: 5
Duration: 1 Hour

Total Marks:100

Experiments

Code of Conduct:

I, Girija B / 192311044, certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Girija B

1 Write a C program to check whether a given string belongs to the language defined by a Context Free Grammar (CFG)

$S \rightarrow 0A1$ $A \rightarrow 0A \mid 1A \mid \epsilon$

C Program

```
#include <stdio.h>

#include <string.h>

int main()
{
    char str[100];
    int n;

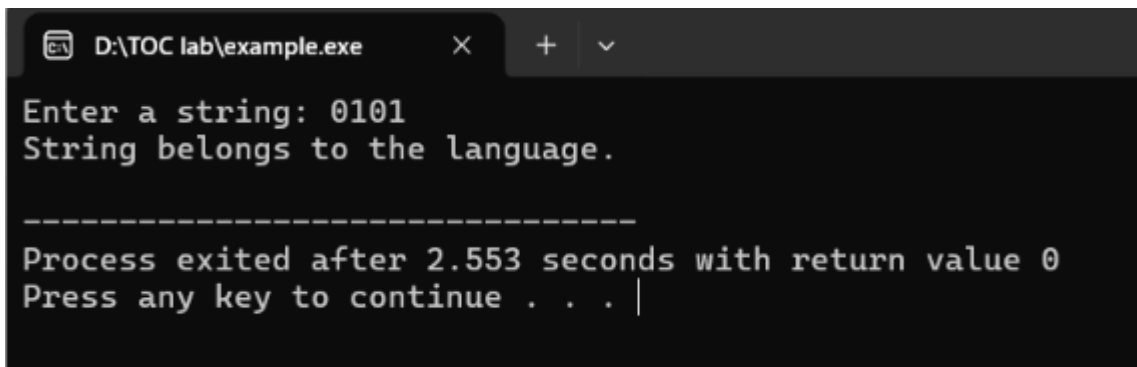
    printf("Enter a string: ");
    scanf("%s", str);

    n = strlen(str);

    if (n >= 2 && str[0] == '0' && str[n - 1] == '1')
        printf("String belongs to the language.\n");
    else
        printf("String does not belong to the language.\n");

    return 0;
}
```

Output:



```
D:\TOC lab\example.exe
Enter a string: 0101
String belongs to the language.

-----
Process exited after 2.553 seconds with return value 0
Press any key to continue . . . |
```

Write a C program to check whether a given string belongs to the language defined by a Context Free Grammar (CFG)

$S \rightarrow 0S0 \mid 1S1 \mid 0 \mid 1 \mid \epsilon$

CODE:

```
#include <stdio.h>
#include <string.h>

int main()
{
    char str[100];
    int i, n, flag = 1;

    printf("Enter a string: ");
    scanf("%s", str);

    n = strlen(str);

    for(i = 0; i < n / 2; i++)
    {
        if(str[i] != str[n - i - 1])
        {
            flag = 0;
            break;
        }
    }

    if(flag == 1)
        printf("String belongs to the language.\n");
    else
```

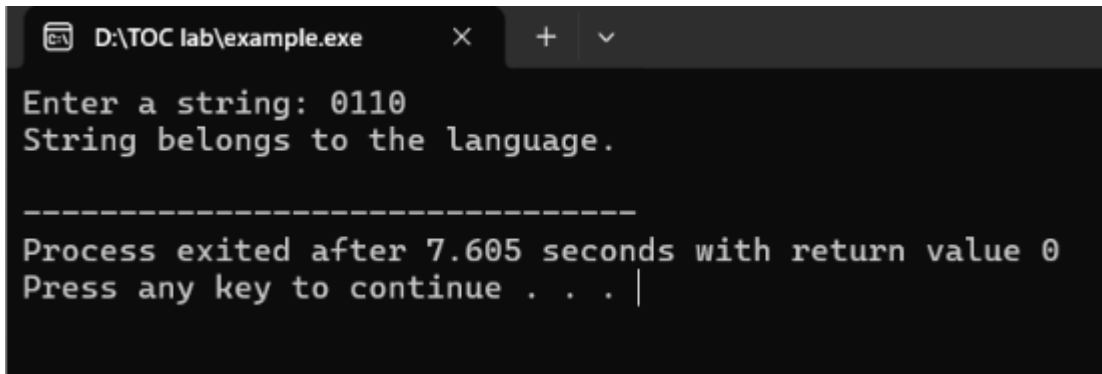
```

        printf("String does not belong to the language.\n");

    return 0;
}

```

Output:



```

D:\TOC lab\example.exe
Enter a string: 0110
String belongs to the language.

-----
Process exited after 7.605 seconds with return value 0
Press any key to continue . . . |

```

3. Write a C program to check whether a given string belongs to the language defined by a Context Free Grammar (CFG)

$S \rightarrow OS0 \mid A \ A \rightarrow 1A \mid \epsilon$

CODE:

```
#include <stdio.h>
```

```
#include <string.h>
```

```
int main()
```

```
{
```

```
    char str[100];
```

```
    int n, i, j, valid = 1;
```

```
    printf("Enter a string: ");
```

```
    scanf("%s", str);
```

```
    n = strlen(str);
```

```
    i = 0;
```

```

j = n - 1;

/* Check equal number of 0s at both ends */
while (i < j && str[i] == '0' && str[j] == '0')
{
    i++;
    j--;
}

/* Middle part must contain only 1s */
while (i <= j)
{
    if (str[i] != '1')
    {
        valid = 0;
        break;
    }
    i++;
}

if (valid)
    printf("String belongs to the language.\n");
else
    printf("String does not belong to the language.\n");

getchar();
getchar();

return 0;
}

```

OUTPUT:

```
D:\TOC lab\example.exe  X + v
Enter a string: 0110
String belongs to the language.

-----
Process exited after 7.605 seconds with return value 0
Press any key to continue . . . |
```

Write a C program to check whether a given string belongs to the language defined by a Context Free Grammar (CFG)

$S \rightarrow 0S1 \mid \epsilon$

Code:

```
#include <stdio.h>
```

```
#include <string.h>
```

```
int main()
```

```
{
```

```
    char str[100];
```

```
    int i, n, zeros = 0, ones = 0, valid = 1;
```

```
    printf("Enter a string: ");
```

```
    scanf("%s", str);
```

```
    n = strlen(str);
```

```
    /* Count 0s first */
```

```
    i = 0;
```

```
    while (i < n && str[i] == '0')
```

```
    {
```

```
        zeros++;
```

```
        i++;
```

```

}

/* Count 1s after all 0s */
while (i < n && str[i] == '1')
{
    ones++;
    i++;
}

/* Check equal number of 0s and 1s and correct order */
if (i != n || zeros != ones)
    valid = 0;

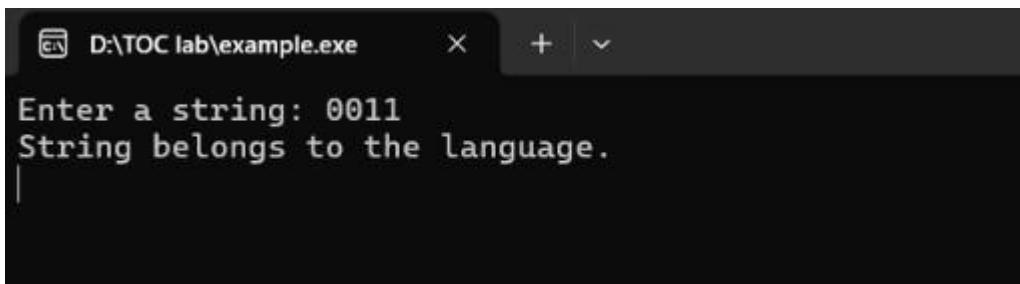
if (valid)
    printf("String belongs to the language.\n");
else
    printf("String does not belong to the language.\n");

getchar();
getchar();

return 0;
}

```

OUTPUT:



```

D:\TOC lab\example.exe
Enter a string: 0011
String belongs to the language.
|

```

5. Write a C program to check whether a given string belongs to the language defined by a Context Free Grammar (CFG)

$S \rightarrow A101A$ $A \rightarrow 0A \mid 1A \mid \epsilon$

CODE:

```
#include <stdio.h>

#include <string.h>

int main()
{
    char str[100];
    int i, n, found = 0;

    printf("Enter a string: ");
    scanf("%s", str);

    n = strlen(str);

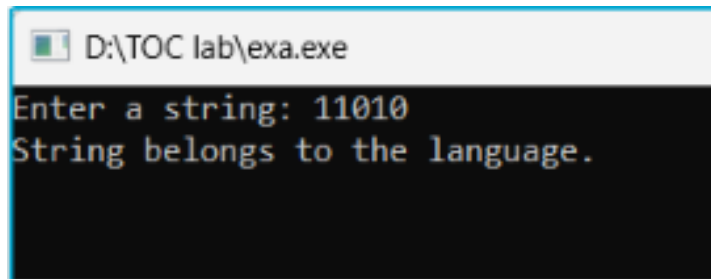
    for(i = 0; i <= n - 3; i++)
    {
        if(str[i] == '1' && str[i + 1] == '0' && str[i + 2] == '1')
        {
            found = 1;
            break;
        }
    }

    if(found)
        printf("String belongs to the language.\n");
    else
        printf("String does not belong to the language.\n");
}
```



```
    getchar();  
    getchar();  
  
    return 0;  
}
```

OUTPUT:

A screenshot of a Windows command prompt window. The title bar at the top reads "D:\TOC lab\exa.exe". The command prompt has a black background with white text. It displays the prompt "Enter a string: " followed by the input "11010". Below this, it displays the output "String belongs to the language.".

```
D:\TOC lab\exa.exe  
Enter a string: 11010  
String belongs to the language.
```